

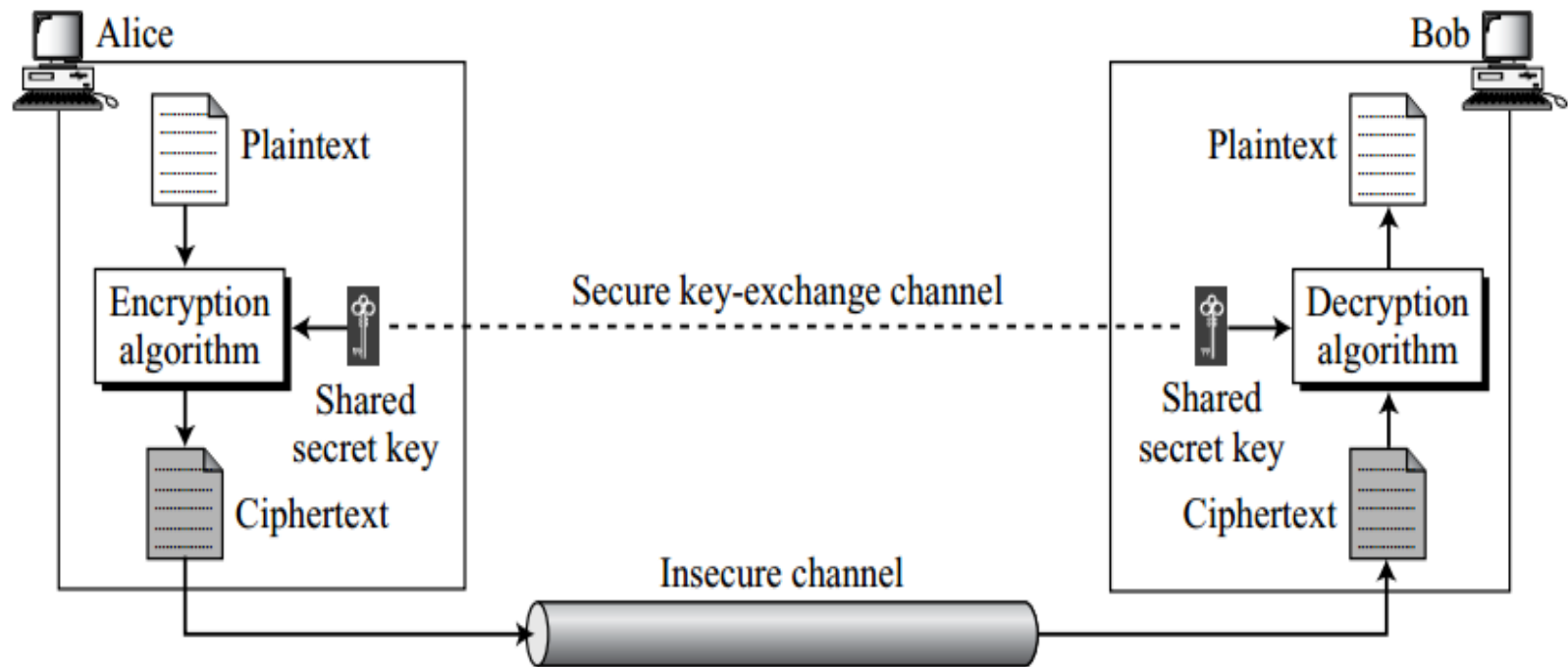
Symmetric-key Cipher

Introduction

- Symmetric-key encipherment is like locking and unlocking with the same key



General idea of symmetric-key cipher



Encryption: $C = E_k(P)$

Decryption: $P = D_k(C)$

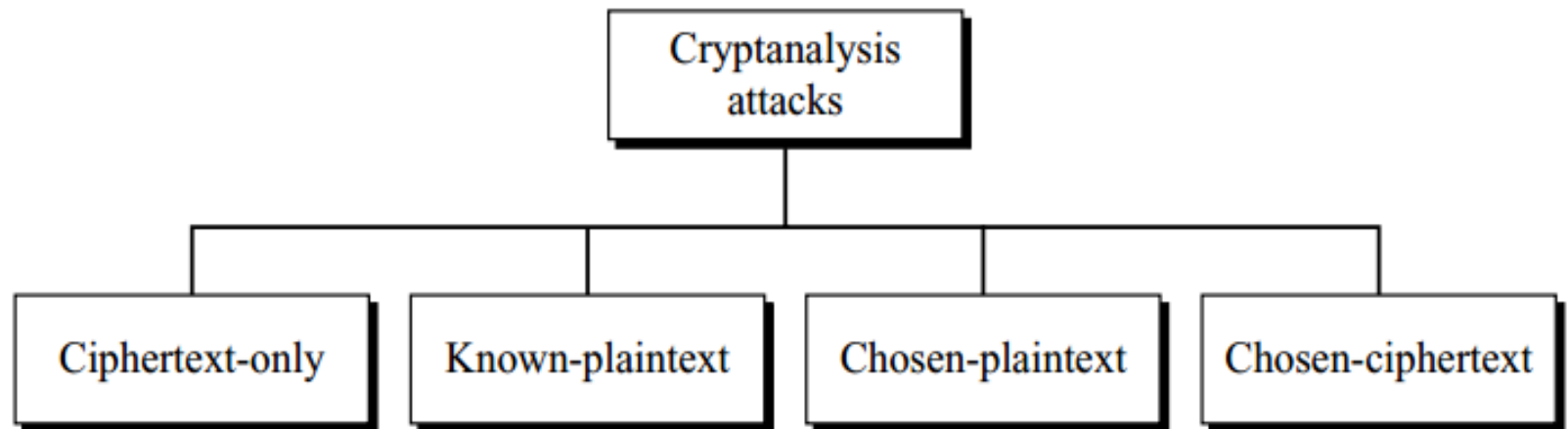
In which, $D_k(E_k(x)) = E_k(D_k(x)) = x$

Concern

- There is an established **secret key** between Alice and Bob.
- Alice and Bob can use the same key for communication (So, **symmetric**)
- **Number of keys** is a concern
 - If there are m people in a group, each person needs $(m - 1)$ keys to communicate with the rest of the group
 - A total $(m \times (m - 1))/2$ keys will be needed in a group

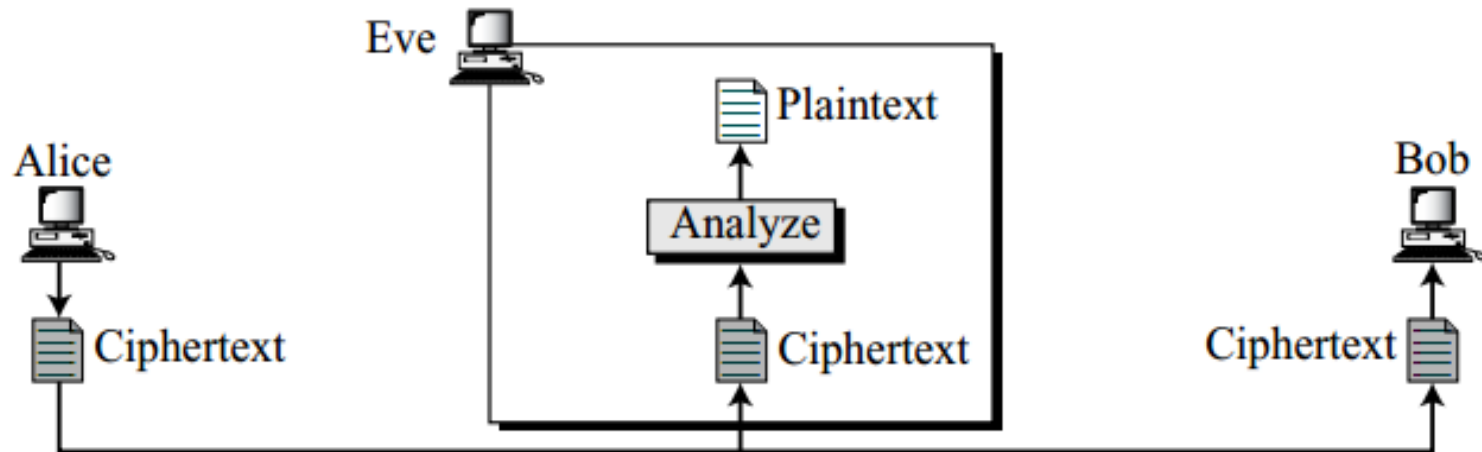
Cryptanalysis

- **Cryptography** is the science and art of creating secret codes
- **Cryptanalysis** is the science and art of breaking those codes
- Cryptanalysis attacks:



Ciphertext-Only Attack

- Attacker has access of some ciphertexts, tries to find the corresponding key and the plaintext.

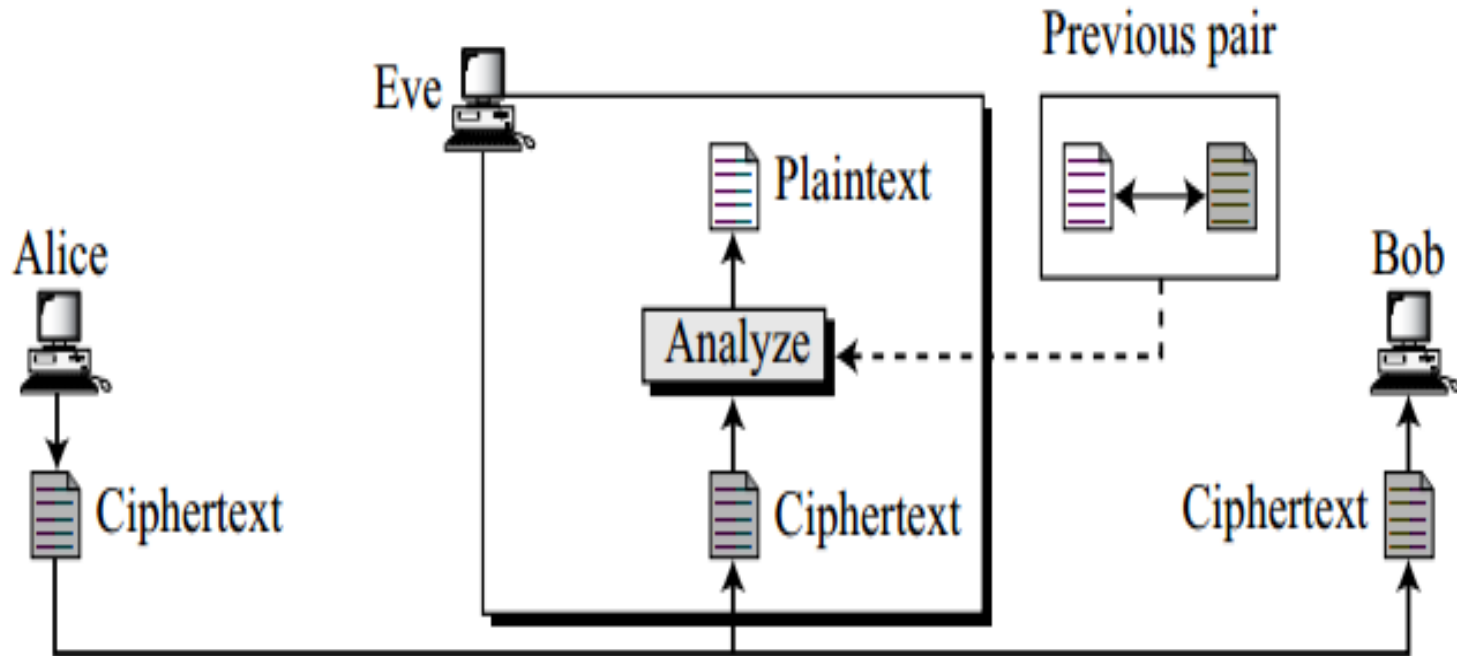


Ciphertext-Only Attack

- Various methods can be used in ciphertext-only attack
 - Brute-Force Attack
 - Statistical Attack
 - Pattern Attack

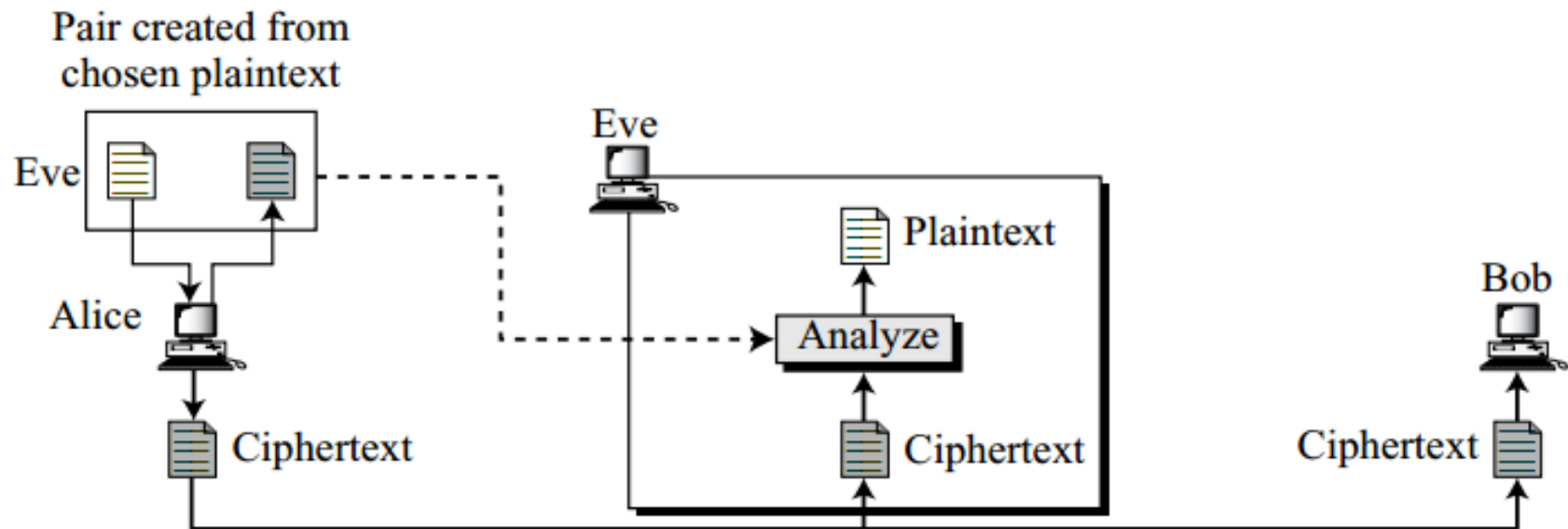
Known Plaintext Attack

- Eve has access to some plaintext/ciphertext pairs



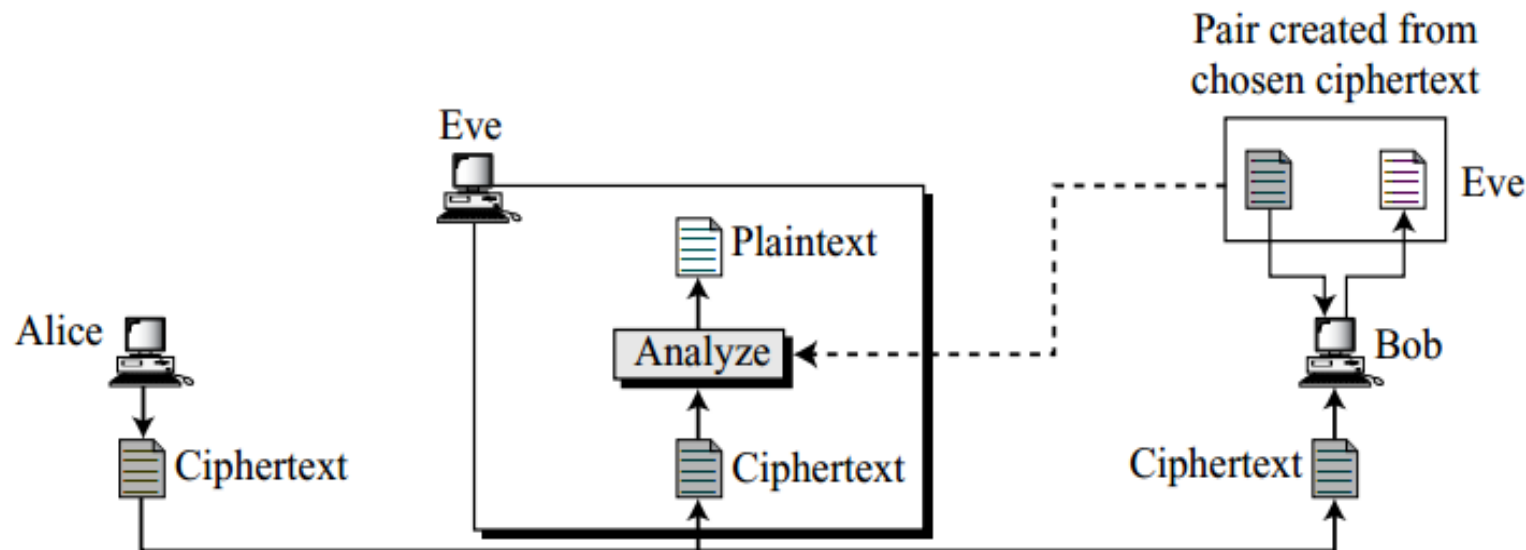
Chosen-Plaintext Attack

- It is similar to the known-plaintext attack, but the plaintext/ ciphertext pairs have been chosen by the attacker



Chosen-Ciphertext Attack

- It is similar to the chosen-plaintext attack
 - except that Eve chooses some ciphertext and decrypts it to form a ciphertext/plaintext pair



Categories of Traditional Ciphers

- Traditional symmetric-key ciphers are of two broad categories:
 - Substitution ciphers
 - one symbol in the CT is replaced with another symbol
 - Transposition ciphers
 - the position of symbols in the plaintext is reordered

Substitution Cipher

- A substitution cipher replaces one symbol with another.
- They are of two types.
 - monoalphabetic ciphers
 - polyalphabetic ciphers

Monoalphabetic Ciphers

- A character (or a symbol) in the plaintext is always changed to the same character (or symbol) in the ciphertext regardless of its position in the text.
- Relationship between letters in the plaintext and the ciphertext is **one-to-one**.

Monoalphabetic Ciphers

- Example

Plaintext: hello Ciphertext: KHOOR
The cipher is probably monoalphabetic

Plaintext: hello Ciphertext: ABNZF
The cipher is not monoalphabetic

List of monoalphabetic cipher

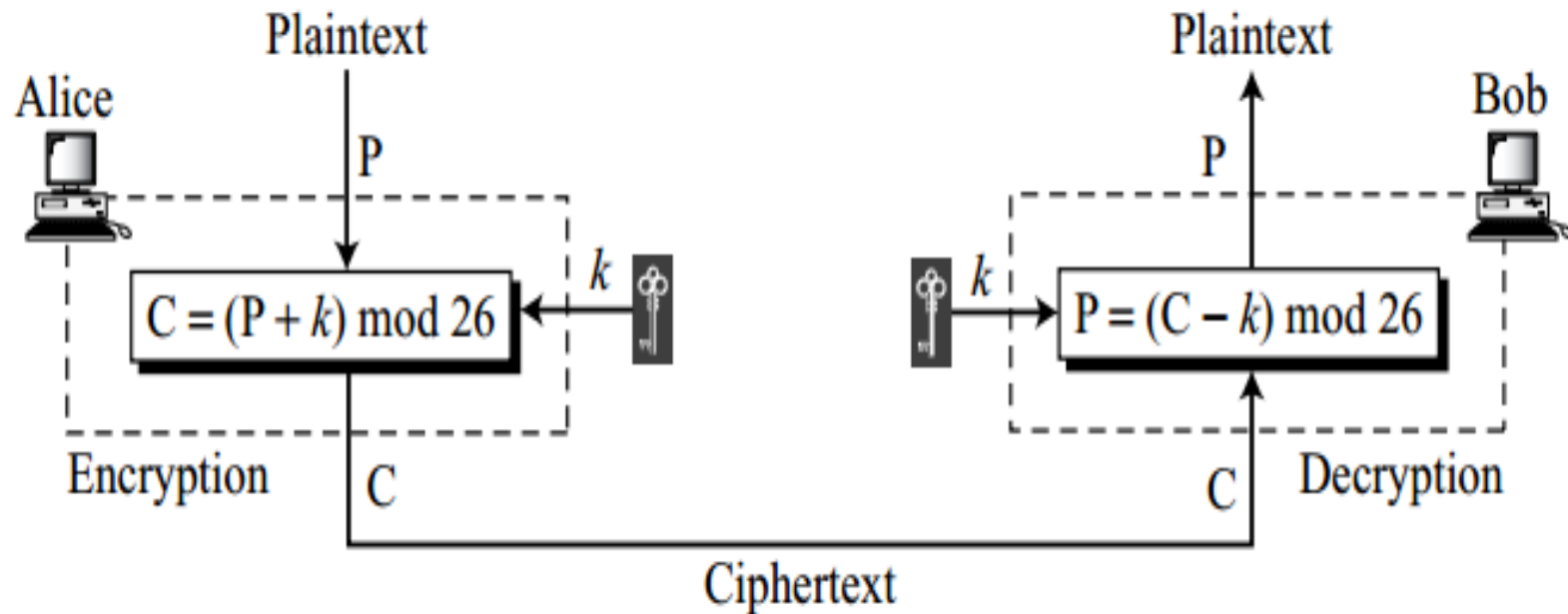
- Additive Cipher
- Multiplicative Ciphers
- Affine Cipher
- Monoalphabetic Substitution Cipher

Additive Cipher

- Simplest, having mathematical nature
- Representation of plaintext and ciphertext characters in Z_{26}
- The plaintext consists of lowercase letters (a to z), the ciphertext consists of uppercase letters (A to Z).

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Additive cipher



$$P_1 = (C - k) \bmod 26 = (P + k - k) \bmod 26 = P$$

When the cipher is additive, the plaintext, ciphertext, and key are integers in $\mathbb{Z}_{26}$.

Example

- Use the additive cipher with key = 15 to encrypt and decrypt the message “hello”.

Plaintext: h → 07	Encryption: $(07 + 15) \bmod 26$	Ciphertext: 22 → W
Plaintext: e → 04	Encryption: $(04 + 15) \bmod 26$	Ciphertext: 19 → T
Plaintext: l → 11	Encryption: $(11 + 15) \bmod 26$	Ciphertext: 00 → A
Plaintext: l → 11	Encryption: $(11 + 15) \bmod 26$	Ciphertext: 00 → A
Plaintext: o → 14	Encryption: $(14 + 15) \bmod 26$	Ciphertext: 03 → D

Ciphertext: W → 22	Decryption: $(22 - 15) \bmod 26$	Plaintext: 07 → h
Ciphertext: T → 19	Decryption: $(19 - 15) \bmod 26$	Plaintext: 04 → e
Ciphertext: A → 00	Decryption: $(00 - 15) \bmod 26$	Plaintext: 11 → l
Ciphertext: A → 00	Decryption: $(00 - 15) \bmod 26$	Plaintext: 11 → l
Ciphertext: D → 03	Decryption: $(03 - 15) \bmod 26$	Plaintext: 14 → o

- Additive ciphers are sometimes referred to as shift ciphers or Caesar cipher.
- Additive ciphers are vulnerable to ciphertext-only attacks using exhaustive key searches (**brute-force attacks**).
- One of the keys, **zero**, is useless (the ciphertext is the same as the plaintext). This leaves only 25 possible keys.
- Additive ciphers are also subject to statistical attacks

Ciphertext: UVACLYFZLJBYL

K = 1	→	Plaintext: tuzbkxeykiaxk
K = 2	→	Plaintext: styajwdxjhzwj
K = 3	→	Plaintext: rsxzivcwigyvi
K = 4	→	Plaintext: qrwyhubvhfxuh
K = 5	→	Plaintext: pqvxgtaugewtg
K = 6	→	Plaintext: opuwfsztfdvsvf
K = 7	→	Plaintext: notverysecure

Eve tries keys from 1 to 7. With a key of 7, the plaintext is “not very secure”, which makes sense.

Eve has intercepted the following ciphertext. Using a statistical attack, find the plaintext.

XLILSYWIMWRS AJSVWEPIJSVJSYVQMPPMSRHSPPEVWMXMWASVX-LQSVILY-
VVCFIJSVIXLIWIPPIVVIGIMZIWQSVISJJIVW

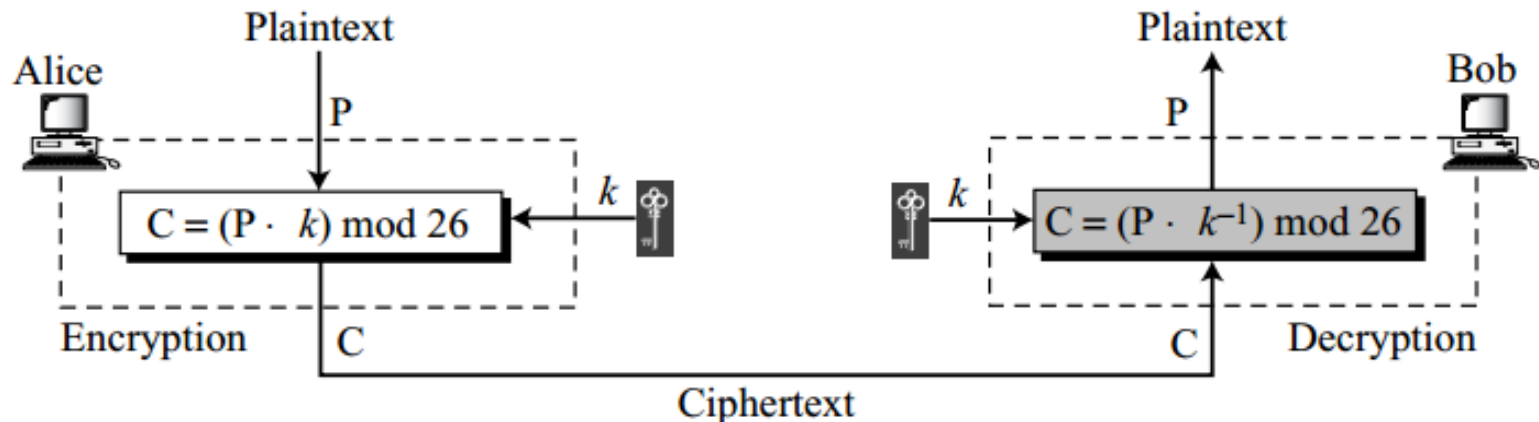
Solution

When Eve tabulates the frequency of letters in this ciphertext, she gets: I =14, V =13, S =12, and so on. The most common character is I with 14 occurrences. This shows that character I in the ciphertext probably corresponds to the character e in plaintext. This means $\text{key} = 4$. Eve deciphers the text to get

the house is now for sale for four million dollars it is worth more hurry before the seller
receives more offers

Multiplicative Ciphers

- The encryption algorithm specifies multiplication of the plaintext by the key and the decryption algorithm specifies division of the ciphertext by the key



In a multiplicative cipher, the plaintext and ciphertext are integers in $\mathbb{Z}_{26}$; the key is an integer in $\mathbb{Z}_{26}^*$.

Example

The key needs to be in $\mathbf{Z}_{26}^*$. This set has only 12 members: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25.

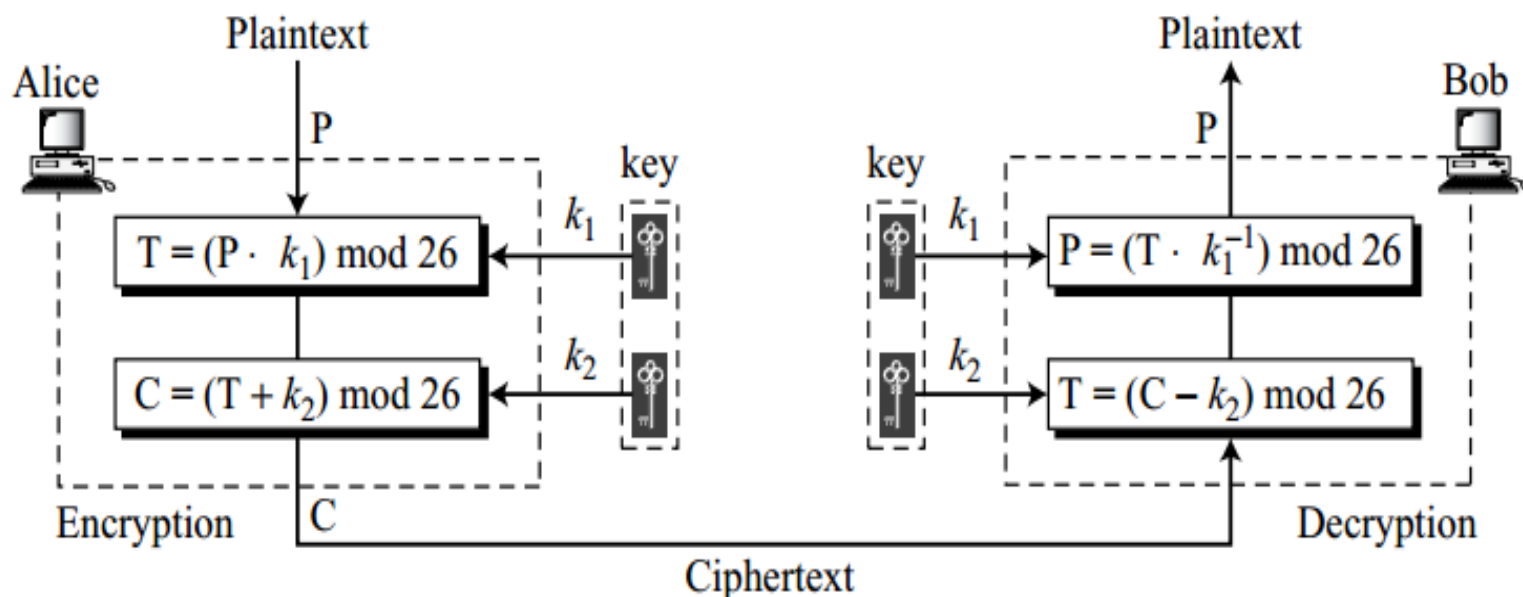
We use a multiplicative cipher to encrypt the message “hello” with a key of 7. The ciphertext is “XCZZU”.

Plaintext: h $\rightarrow$ 07	Encryption: $(07 \times 07) \bmod 26$	ciphertext: 23 $\rightarrow$ X
Plaintext: e $\rightarrow$ 04	Encryption: $(04 \times 07) \bmod 26$	ciphertext: 02 $\rightarrow$ C
Plaintext: l $\rightarrow$ 11	Encryption: $(11 \times 07) \bmod 26$	ciphertext: 25 $\rightarrow$ Z
Plaintext: l $\rightarrow$ 11	Encryption: $(11 \times 07) \bmod 26$	ciphertext: 25 $\rightarrow$ Z
Plaintext: o $\rightarrow$ 14	Encryption: $(14 \times 07) \bmod 26$	ciphertext: 20 $\rightarrow$ U

Affine Cipher

- It is a combination of additive and multiplicative ciphers with a pair of keys
- The first key is used with the multiplicative cipher; the second key is used with the additive cipher.
- We should be sure that each one has an inverse at the other side of the line and that they are used in reverse order in the encryption and decryption.

Affine Cipher



In the affine cipher, the relationship between the plaintext P and the ciphertext C is

$$C = (P \times k_1 + k_2) \bmod 26$$

$$P = ((C - k_2) \times k_1^{-1}) \bmod 26$$

where k_1^{-1} is the multiplicative inverse of k_1 and $-k_2$ is the additive inverse of k_2

Example

Use an affine cipher to encrypt the message “hello” with the key pair (7, 2).

Solution

We use 7 for the multiplicative key and 2 for the additive key. We get “ZEBBW”.

P: h $\rightarrow$ 07	Encryption: $(07 \times 7 + 2) \bmod 26$	C: 25 $\rightarrow$ Z
P: e $\rightarrow$ 04	Encryption: $(04 \times 7 + 2) \bmod 26$	C: 04 $\rightarrow$ E
P: l $\rightarrow$ 11	Encryption: $(11 \times 7 + 2) \bmod 26$	C: 01 $\rightarrow$ B
P: l $\rightarrow$ 11	Encryption: $(11 \times 7 + 2) \bmod 26$	C: 01 $\rightarrow$ B
P: o $\rightarrow$ 14	Encryption: $(14 \times 7 + 2) \bmod 26$	C: 22 $\rightarrow$ W

Affine Cipher

- The additive cipher is a special case of an affine cipher in which $k_1 = 1$.
- The multiplicative cipher is a special case of affine cipher in which $k_2 = 0$.

Monoalphabetic Substitution Cipher

- The additive, multiplicative, and affine ciphers have small key domains (vulnerable to brute-force attack)
- A better solution is to create a mapping between each plaintext character and the corresponding ciphertext character.

- An example key for monoalphabetic substitution cipher

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	N	O	A	T	R	B	E	C	F	U	X	D	Q	G	Y	L	K	H	V	I	J	M	P	Z	S	W

Encrypt the message

this message is easy to encrypt but hard to find the key

The ciphertext is

ICFVQRVVNEFVRNVSIYRGAHSLIOJICNHTIYBFGTICRXRS

- The size of the key space for the monoalphabetic substitution cipher is $26!$
- The monoalphabetic ciphers do not change the frequency of characters in the ciphertext, which makes the ciphers vulnerable to **statistical attack**.
- Solution

Polyalphabetic Ciphers

Polyalphabetic Ciphers

- Each occurrence of a character may have a different substitute.
- Relationship between a character in the plaintext to a character in the cipher text is one-to-many.
- It has advantage of hiding the letter frequency
- key should be a stream of subkeys.
- $k = (k_1, k_2, k_3, \dots)$ in which k_i is used to encipher the i th character in the plaintext to create the i th character in the ciphertext.

Autokey Cipher

- The subkeys are automatically created from the plaintext cipher characters during the encryption process
- The key is a stream of subkeys, each subkey is used to encrypt the corresponding character in the plaintext.
- The first subkey is a predetermined value, second subkey is the value of the first plaintext character, third subkey is the value of the second plaintext

Autokey Cipher

$$P = P_1 P_2 P_3 \dots$$

$$C = C_1 C_2 C_3 \dots$$

$$k = (k_1, P_1, P_2, \dots)$$

$$\text{Encryption: } C_i = (P_i + k_i) \bmod 26$$

$$\text{Decryption: } P_i = (C_i - k_i) \bmod 26$$

Example : Autokey cipher with initial key value $k_1 = 12$

Plaintext:	a	t	t	a	c	k	i	s	t	o	d	a	y
P's Values:	00	19	19	00	02	10	08	18	19	14	03	00	24
Key stream:	12	00	19	19	00	02	10	08	18	19	14	03	00
C's Values:	12	19	12	19	02	12	18	00	11	7	17	03	24
Ciphertext:	M	T	M	T	C	M	S	A	L	H	R	D	Y

Autokey Cipher

- **Cryptanalysis**
 - hides the single-letter frequency statistics of the plaintext
 - vulnerable to the brute-force attack
 - The first subkey can be only one of the 25 values
- **Need**
 - Polyalphabetic ciphers that hides the characteristics of the language
 - Should have **large key domains**.

Playfair Cipher

- The secret key in this cipher is made of 25 alphabet letters arranged in a 5×5 matrix
- letters **I** and **J** are considered as the same when encrypting.
- Different arrangements of the letters in the matrix can create many different secret keys.

Secret Key =

L	G	D	B	A
Q	M	H	E	C
U	R	N	I/J	F
X	V	S	O	K
Z	Y	W	T	P

Playfair Cipher

- Before encryption, if the two letters in a pair are the same, a bogus letter is inserted to separate them.
- If the number of characters in the plaintext is odd, one extra bogus character is added at the end to make the number of characters even.
- The key is a stream of subkeys in which the subkeys are created two at a time.

Playfair Cipher

- The cipher uses three rules for encryption:
 - a. If the two letters in a pair are located in the same row of the secret key, the corresponding encrypted character for each letter is the next letter to the right in the same row (with wrapping to the beginning of the row if the plaintext letter is the last character in the row).
 - b. If the two letters in a pair are located in the same column of the secret key, the corresponding encrypted character for each letter is the letter beneath it in the same column (with wrapping to the beginning of the column if the plaintext letter is the last character in the column).
 - c. If the two letters in a pair are not in the same row or column of the secret, the corresponding encrypted character for each letter is a letter that is in its own row but in the same column as the other letter.

Playfair Cipher

$P = P_1P_2P_3 \dots$

$C = C_1C_2C_3\dots$

$k = [(k_1, k_2), (k_3, k_4), \dots]$

Encryption: $C_i = k_i$

Decryption: $P_i = k_i$

Example: encrypt the plaintext “hello” using the key

L	G	D	B	A
Q	M	H	E	C
U	R	N	I/J	F
X	V	S	O	K
Z	Y	W	T	P

he → EC

lx → QZ

lo → BX

Plaintext: hello

Ciphertext: ECQZBX

Vignere Cipher

- The key stream is a repetition of an initial secret key stream of length m , where $1 \leq m \leq 26$.

$$P = P_1 P_2 P_3 \dots$$

$$C = C_1 C_2 C_3 \dots$$

$$K = [(k_1, k_2, \dots, k_m), (k_1, k_2, \dots, k_m), \dots]$$

$$\text{Encryption: } C_i = P_i + k_i$$

$$\text{Decryption: } P_i = C_i - k_i$$

- Vignere key stream does not depend on the plaintext characters.
- The key stream can be created without knowing the plaintext.

Vigenere Cipher

- Example: encrypt the message “She is listening” using the 6-character keyword “PASCAL”.
- The initial key stream is (15, 0, 18, 2, 0, 11)
- key stream is the repetition of this initial key stream (as many times as needed).

Plaintext:	s	h	e	i	s	l	i	s	t	e	n	i	n	g
P's values:	18	07	04	08	18	11	08	18	19	04	13	08	13	06
Key stream:	<i>15</i>	<i>00</i>	<i>18</i>	<i>02</i>	<i>00</i>	<i>11</i>	<i>15</i>	<i>00</i>	<i>18</i>	<i>02</i>	<i>00</i>	<i>11</i>	<i>15</i>	<i>00</i>
C's values:	07	07	22	10	18	22	23	18	11	6	13	19	02	06
Ciphertext:	H	H	W	K	S	W	X	S	L	G	N	T	C	G

Hill Cipher

- The plaintext is divided into equal-size blocks.
- The blocks are encrypted one at a time in such a way that each character in the block contributes to the encryption of other characters in the block.
- For this reason, the Hill cipher belongs to a category of ciphers called **block ciphers**.

Hill Cipher

- The key is a square matrix of size $m \times m$ in which m is the size of the block.
- The key matrix in the Hill cipher needs to have a **multiplicative inverse**.
- Key in the Hill cipher

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1m} \\ k_{21} & k_{22} & \dots & k_{2m} \\ \vdots & \vdots & & \vdots \\ k_{m1} & k_{m2} & \dots & k_{mm} \end{bmatrix}$$

Hill Cipher

- Suppose, there are m characters in the plaintext block $P_1, P_2, \dots, P_m$
- The corresponding characters in the cipher text block are $C_1, C_2, \dots, C_m$.

$$\begin{aligned}C_1 &= P_1 k_{11} + P_2 k_{21} + \dots + P_m k_{m1} \\C_2 &= P_1 k_{12} + P_2 k_{22} + \dots + P_m k_{m2} \\&\dots \\C_m &= P_1 k_{1m} + P_2 k_{2m} + \dots + P_m k_{mm}\end{aligned}$$

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1m} \\ k_{21} & k_{22} & \dots & k_{2m} \\ \vdots & \vdots & & \vdots \\ k_{m1} & k_{m2} & \dots & k_{mm} \end{bmatrix}$$

key needs to be selected carefully. Bob will not be able to decrypt the ciphertext sent by Alice if the matrix does not have a multiplicative inverse.

Example

- The plaintext is **code is ready**
- We can make a 3×4 matrix when adding extra bogus character “z” to the last block and removing the spaces. Number of blocks is 3
- The ciphertext is “**OHKNIHGKLISS**”.
- Bob can decrypt the message using the inverse of the key matrix.

$$\begin{array}{c} \mathbf{C} \\ \left[\begin{array}{cccc} 14 & 07 & 10 & 13 \\ 08 & 07 & 06 & 11 \\ 11 & 08 & 18 & 18 \end{array} \right] \end{array} = \begin{array}{c} \mathbf{P} \\ \left[\begin{array}{cccc} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{array} \right] \end{array} \begin{array}{c} \mathbf{K} \\ \left[\begin{array}{cccc} 09 & 07 & 11 & 13 \\ 04 & 07 & 05 & 06 \\ 02 & 21 & 14 & 09 \\ 03 & 23 & 21 & 08 \end{array} \right] \end{array}$$

a. Encryption

$$\begin{array}{c} \mathbf{P} \\ \left[\begin{array}{cccc} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{array} \right] \end{array} = \begin{array}{c} \mathbf{C} \\ \left[\begin{array}{cccc} 14 & 07 & 10 & 13 \\ 08 & 07 & 06 & 11 \\ 11 & 08 & 18 & 18 \end{array} \right] \end{array} \begin{array}{c} \mathbf{K}^{-1} \\ \left[\begin{array}{cccc} 02 & 15 & 22 & 03 \\ 15 & 00 & 19 & 03 \\ 09 & 09 & 03 & 11 \\ 17 & 00 & 04 & 07 \end{array} \right] \end{array}$$

b. Decryption

TRANSPOSITION CIPHERS

Transposition Cipher

- A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.
- It reorders (**transposes**) the symbols.
- Two types:
 - Keyless Transposition Ciphers
 - Keyed Transposition Ciphers

Keyless Transposition Ciphers

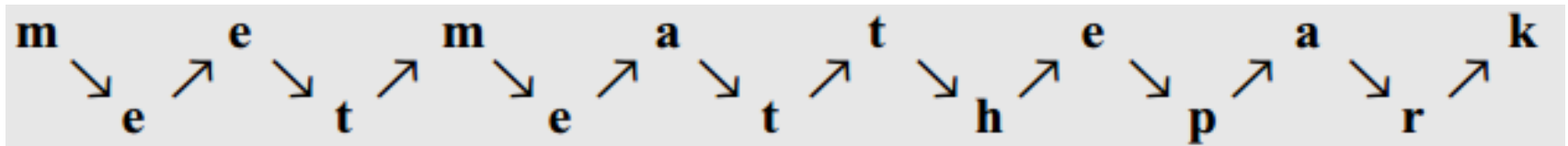
- There are two methods for permutation of characters.
 - the text is written into a table column by column and then transmitted row by row.
 - The text is written into the table row by row and then transmitted column by column.
- Example: Rail fence cipher

Rail fence cipher

- It is a keyless transposition cipher in which the text is written into a table column by column and then transmitted row by row.
- The plaintext is arranged in two lines as a zigzag pattern (which means column by column).
- The ciphertext is created reading the pattern row by row.

Rail fence cipher

- To send the message “Meet me at the park” to Bob, Alice writes



ciphertext “MEMATEAKETETHPR”

- The receiver divides it in half.
- The first half forms the first row; the second half, the second row. Then reads the result in zigzag

There is no key and the number of rows is fixed (2), the cryptanalysis of the ciphertext would be very easy

Rail fence cipher

- Example: Alice and Bob can agree on the number of columns and use the second method.
- Alice writes the same plaintext, row by row, in a table of four columns.

m	e	e	t
m	e	a	t
t	h	e	p
a	r	k	

She then creates the ciphertext “MMTAEEHREAEKTP”

- Bob receives the ciphertext and follows the reverse process.

- He writes the received message, column by column, and reads it row by row as the plaintext.

Vulnerable if the number of columns is intercepted by attacker

Keyed Transposition Ciphers

- The keyless ciphers permute the characters by using writing plaintext in one way (row by row) and reading it in another way (column by column).
- The permutation is done on the whole plaintext to create the whole ciphertext.
- Another method is to divide the plaintext into groups of predetermined size, called **blocks**, and then use a key to permute the characters in each block separately.

Keyed Transposition Ciphers

- **Example**

Alice needs to send the message “Enemy attacks tonight” to Bob.

- Alice and Bob have agreed to divide the text into **groups of five characters** and then permute the characters in each group.
- Alice and Bob used the following key:

Encryption ↓	<table><tr><td>3</td><td>1</td><td>4</td><td>5</td><td>2</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr></table>	3	1	4	5	2	1	2	3	4	5	↑ Decryption
3	1	4	5	2								
1	2	3	4	5								

Keyed Transposition Ciphers

- Grouping after adding a bogus character at the end

e n e m y a t t a c k s t o n i g h t z

Encryption ↓

3	1	4	5	2
1	2	3	4	5

↑ Decryption

The permutation yields

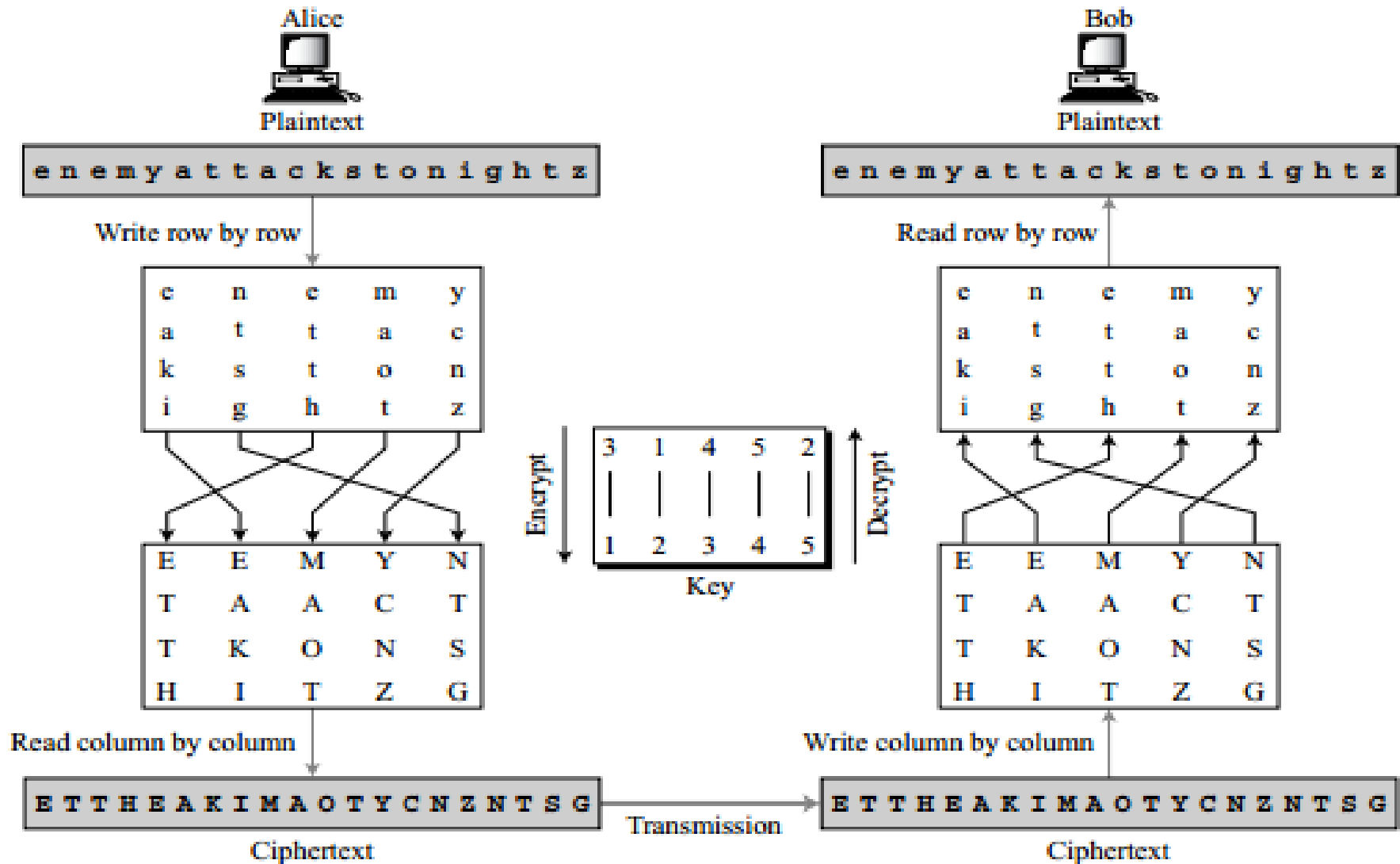
E E M Y N T A A C T T K O N S H I T Z G

Bob divides the ciphertext into 5-character groups and, using the key in the reverse order, finds the plaintext.

Combining Two Approaches

- More recent transposition ciphers combine the two approaches to achieve better.
- Encryption or decryption is done in three steps.
 - the text is written into a table row by row.
 - the permutation is done by reordering the columns.
 - the new table is read column by column.
- These types of ciphers are often referred to as **keyed columnar transposition ciphers**

Combining Two Approaches



Using Matrices in Transposition Ciphers

- The plaintext and ciphertext are $l \times m$ matrices representing the numerical values of the characters
- The keys are square matrices of size $m \times m$
- In a permutation matrix, every row or column has exactly one 1 and the rest of the values are 0s.
- Encryption is performed by multiplying the plaintext matrix by the key matrix to get the ciphertext matrix
- Decryption is performed by multiplying the ciphertext by the inverse key matrix to get the plaintext matrix.

Using Matrices in Transposition Ciphers

- Example

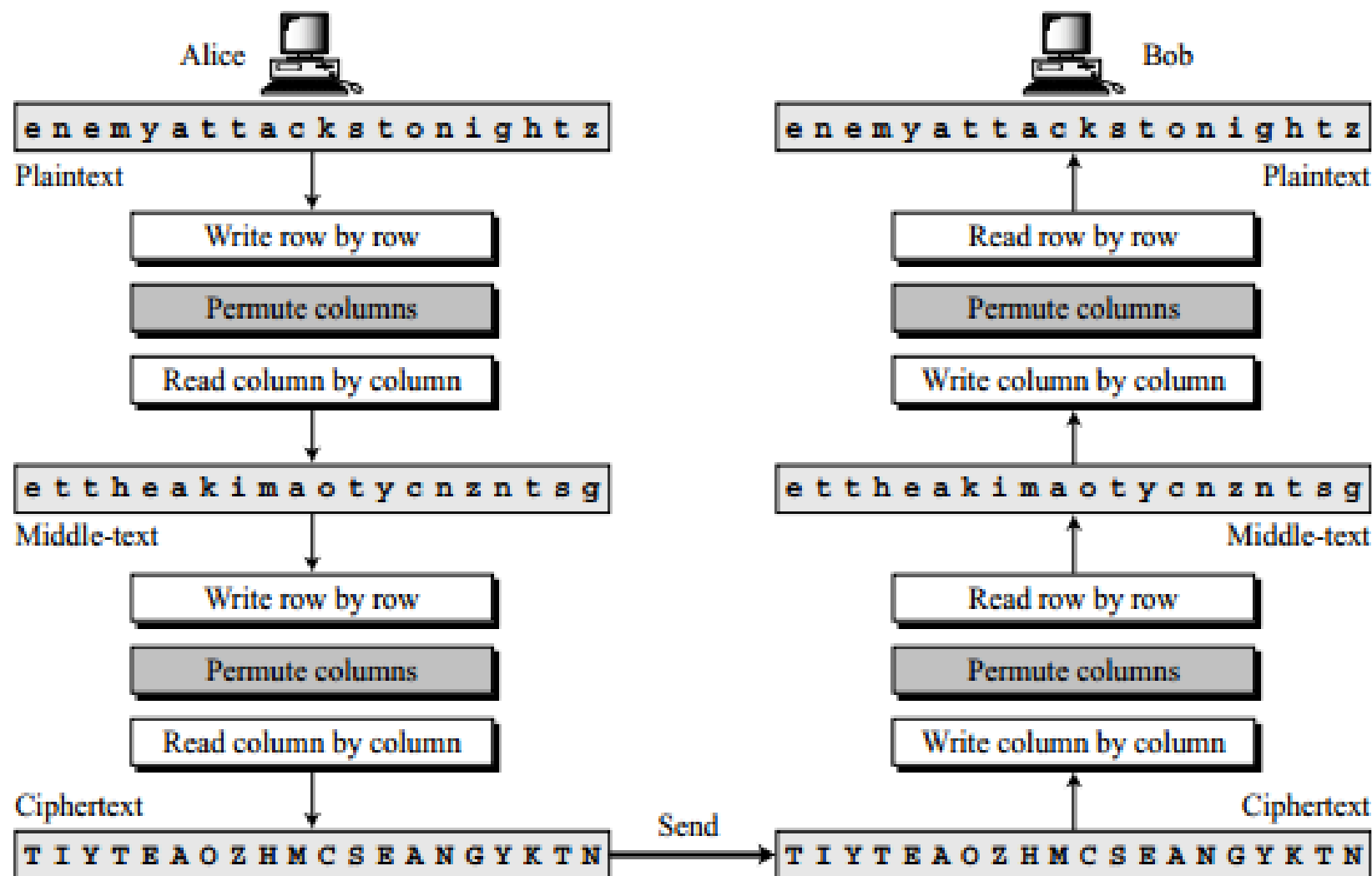
Multiplying the 4×5 plaintext matrix by the 5×5 encryption key gives the 4×5 ciphertext matrix.

$$\begin{array}{c}
 \begin{bmatrix} 04 & 13 & 04 & 12 & 24 \\ 00 & 19 & 19 & 00 & 02 \\ 10 & 18 & 19 & 14 & 13 \\ 08 & 06 & 07 & 19 & 25 \end{bmatrix} \\
 \text{Plaintext}
 \end{array}
 \cdot
 \begin{array}{c}
 \begin{array}{ccccc}
 3 & 1 & 4 & 5 & 2
 \end{array} \\
 \begin{array}{ccccc}
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow
 \end{array} \\
 \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \\
 \text{Encryption key}
 \end{array}
 =
 \begin{array}{c}
 \begin{bmatrix} 04 & 04 & 12 & 24 & 13 \\ 19 & 00 & 00 & 02 & 19 \\ 19 & 10 & 14 & 13 & 18 \\ 07 & 08 & 19 & 25 & 06 \end{bmatrix} \\
 \text{Ciphertext}
 \end{array}$$

Double Transposition Ciphers

- It can make the job of the cryptanalyst difficult
- Repeats twice the algorithm used for encryption and decryption.
- A different/same key can be used in each step.

Double Transposition Ciphers



Stream Ciphers

- In a stream cipher, encryption and decryption are done one symbol (such as a character or a bit) at a time.
- We have a plaintext stream (P), a ciphertext stream (C), and a key stream (K).

$$P = P_1 P_2 P_3, \dots$$

$$C = C_1 C_2 C_3, \dots$$

$$K = (k_1, k_2, k_3, \dots)$$

$$C_1 = E_{k_1}(P_1)$$

$$C_2 = E_{k_2}(P_2)$$

$$C_3 = E_{k_3}(P_3) \dots$$

Block Ciphers

- A group of plaintext symbols of size m ($m > 1$) are encrypted together creating a group of ciphertext of the same size.
- A single key is used to encrypt the whole block.
- In a block cipher, a ciphertext block depends on the whole plaintext block.
- Playfair ciphers, Hill ciphers are block ciphers
- Every block cipher is a polyalphabetic cipher